
Algorithmic Methods for Mathematical Models

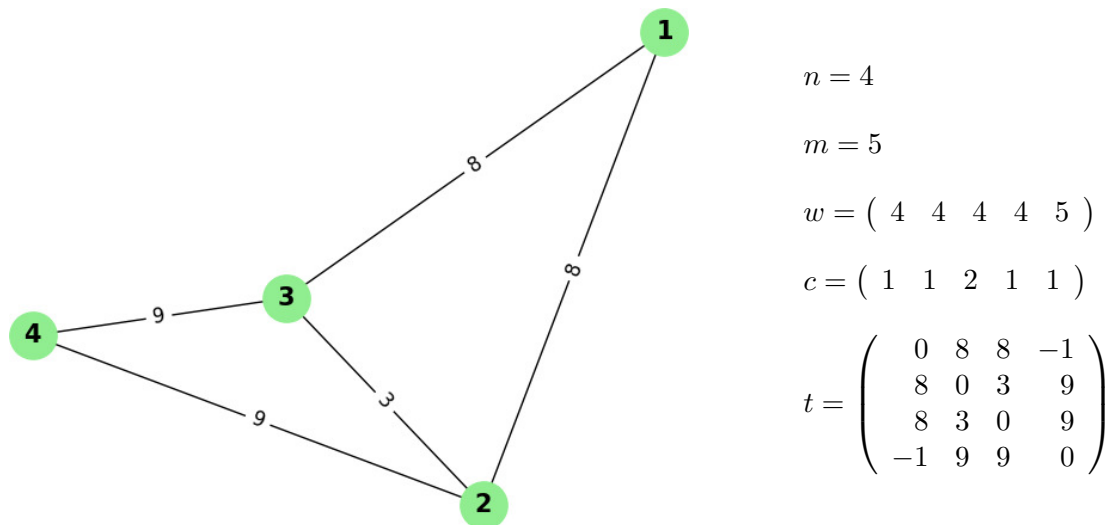
– COURSE PROJECT –

Earth has been invaded by Martians. They have nearly conquered all the planet. The only hope for mankind is the Resistance, who are fighting the enemy with guerrilla warfare. Their next target is to attack the network of fuel distribution of Martians.

Martians have n military bases, which are indexed from 1 to n . Some of these bases are connected with huge pipes that transport fuel. For the time being the Martian network is connected, that is, Martians can send fuel through the pipes from any base to any other base (possibly passing through intermediate bases). The goal of the Resistance is to identify which pipes should be destroyed so that the network becomes disconnected (i.e., it is no longer true that it is connected) and the involved costs are minimized.

More precisely, to destroy pipes the Resistance can hire specialists. There are m of them available, and are indexed from 1 to m . If hired, specialist k ($1 \leq k \leq m$) works exactly w_k hours and charges a fee of c_k k€. Besides, the Computer Science Department of the Resistance has computed a symmetric matrix whose coefficient t_{ij} ($1 \leq i, j \leq n$) is the minimum number of specialist hours required to destroy the pipe between bases i and j . It can be assumed that $t_{ii} = 0$ for all $1 \leq i \leq n$ (and these are the only null values in the matrix), and that $t_{ij} = -1$ if there is no pipe connecting i and j (and these are the only negative values in the matrix). If hired, a specialist can only be sent to destroy one pipe.

Consider the following input data:



For example, if specialists 1, 2, 3 are sent to destroy pipe $\{2, 4\}$ and specialists 4, 5 are sent to destroy pipe $\{3, 4\}$, then the network becomes disconnected (base 4 can no longer be reached from any other base) with cost $1 + 1 + 2 + 1 + 1 = 6$. Note that pipe $\{2, 4\}$ can be destroyed as $4 + 4 + 4 \geq 9$, and pipe $\{3, 4\}$ can be destroyed too since $4 + 5 \geq 9$.

Another solution is to send specialists 1, 2 to destroy pipe $\{1, 2\}$, and specialists 4, 5 to destroy pipe $\{1, 3\}$. Again, the network is disconnected (base 1 becomes unreachable), now with cost $1 + 1 + 1 + 1 = 4$. Notice that as $4 + 4 \geq 8$ pipe $\{1, 2\}$ can be destroyed, and since $4 + 5 \geq 8$ pipe $\{1, 3\}$ can be destroyed as well. This solution can be proved to be optimal.

1. Work to be done:

- (a) Build an integer linear programming model for the problem and implement it in OPL. Simplicity and efficiency will be taken into account when grading.
- (b) Because of the complexity of the problem, heuristic algorithms can also be applied. Here we will consider the following:
 - i. a greedy constructive algorithm,
 - ii. a greedy constructive + a local search procedure,
 - iii. GRASP as a meta-heuristic algorithm. You can reuse the local search procedure that you developed in the previous step.

Design the three algorithms and implement them in the programming language you prefer.

- (c) *Tuning of parameters and instance generation.* Given an instance of input to the problem, the value of m is the *size* of the instance.
 - i. Implement an instance generator that produces random instances for a given size.
 - ii. Tune the α parameter of the GRASP constructive phase with a set of randomly generated instances of large enough size.
 - iii. Generate problem instances with increasingly larger size. Solving each instance with CPLEX should take from 1 to 30 min.
- (d) Compare the performance of CPLEX with the heuristic algorithms, both in terms of computation time and of quality of the solutions as a function of the size of the instances.
- (e) Prepare a report and a presentation of your work on the project.

2. Report:

Prepare a report (8-10 pages) in PDF format including:

- The formal problem statement.
- The integer linear programming model, with a definition and a short description of the variables, the objective function and the constraints. Do not include OPL code in the document, but rather their mathematical formulation.
- For the meta-heuristics, the pseudo-code of your constructive, local search, and GRASP algorithms, including equations for describing the greedy cost function(s) and the RCL.
- Tables or graphs with the tuning of parameters, and with the comparative results.

Together with the report, you should give all sources (OPL code, programs of the meta-heuristics, instances, generator) and instructions on how to use them, so that results can be reproduced.

3. Presentation:

You are expected to make a presentation of your work at the end of the course (at most 10 minutes long; overtime presentations will be terminated). All group members must participate

in the presentation and know all the work in the project. The slots of {26, 29}/05/26 will be devoted to these presentations. The schedule will be announced in its due time.

The slides of the presentation and the report in PDF format should be delivered by **24/05/26**.

The slides should include the full algebraic model, the pseudocode of the algorithms, as well as any other important mathematical equation. As for the results, we expect you to present graphs comparing the performance of the ILP model and heuristics in terms of the quality of the solutions and the time to solve a number of increasingly large problem instances that take up to 30min for CPLEX to solve the ILP. Problem instance generation should be also described.

Please, minimize the auxiliary text in the slides to just what is strictly needed to understand the main content (model, algorithms and results). You are expected to give a full explanation of those contents in the presentation.

On the other hand, the report should also contain that explanation in a well-organized way as a text.

The use of LLMs (ChatGPT, Gemini, etc.) for solving the project is not allowed.